

facturing, and deployment – is free from external constraints.

“Today, despite doing all the design work, you still have a dependency on an external fab for advanced nodes. So sovereignty has to include manufacturing, IP, and manpower,” Gopal points out. In other words, sovereignty isn’t a destination achieved by one fab announcement or one tape-out. It’s a systemic capability.

“India is one of the largest consumer markets for semiconductor devices,” Gopal notes. “Almost any commodity you touch today has a brain – a processor inside it. But a decade back we could not say we had expertise in completely indigenously architecting and designing the chip and its peripherals.”

That gap between consumption and capability is precisely what initiatives like Shakti aim to close. And while the lab has built an end-to-end design ecosystem – from architecture to board and software – manufacturing at a national scale still needs to ramp up.

“Shakti is about building that entire ecosystem – from architecture to physical design to board and software – except the manufacturing.”

The invisible choke-points

If India’s semiconductor debate often focuses on fabs and subsidies, the practitioners inside the ecosystem see a different bottleneck: high-speed IP and licensing costs.

“To make a high-performance system-on-chip useful, you need high-speed peripherals like DDR, Ethernet, Bluetooth,” Gopal says. “The cost is exorbitant and very few companies have a patent for them. Probably a seed funding round of a startup is gone for just that one high-speed peripheral where one or two companies have a monopoly.”

This is the kind of detail that rarely makes headlines but defines real-world chip design economics. Before a startup can even think about tape-

RISC-V Knowledge Centre of Excellence

To ensure that the next generation of hardware engineers are trained, Shakti in collaboration with MeitY has set up the RISC-V Knowledge Centre of Excellence. The initiative aims at providing initial technical support for deep tech startups and education initiatives for semiconductor skill development. The annual symposium titled “Digital India RISC-V (DIR-V)” symposium, to be held in July 2026, is the flagship outreach event that allows our academia, industry, and government partners to meet and outline the roadmap for the future. The Shakti team at IIT Madras acknowledges the support provided by the government ministries and agencies to grow the indigenous semiconductor ecosystem.

out, it may have already exhausted its budget on licensing critical IP blocks. The government has streamlined efforts to make IPs available for startups at an affordable cost under the aegis of the Design Linked Initiative (DLI) scheme.

And then there’s the foundry question. “Today you still have to go outside India for advanced nodes, and we, as a research lab, are nobody to them.” It’s not arrogance on their part, simple market reality, highlights Gopal. “There are millions trying to get access to those foundries, so access to advanced nodes itself becomes a major pain point,” explains Gopal.

This isn’t a complaint so much as a structural observation, I must add in order to avoid any confusion. Globally, foundries operate at capacity, prioritising high-volume clients. For emerging ecosystems like India’s, access to advanced tech itself becomes a strategic challenge.

Even if India succeeds in building fabrication capacity, another challenge looms: talent. “It’s not enough if we build the system... we need people to manage and help run it,” Patanjali says.

“If you build a fab, it’s estimated that you need around 1000 people to run it, and these are extremely highly skilled workers who need to know everything about the semiconductor technology node. So how do we generate these next 1000 skilled workers? We need the entire country to work together on this one single mission,” explains Patanjali.

This is perhaps the most under-discussed dimension of India’s semiconductor ambitions. Building fabs is capital-intensive, but running them is talent-intensive. Without a nationwide effort to build specialised curricula and training pipelines, physical infrastructure alone will not deliver sovereignty.

Then there are the technical realities of advanced nodes, because physics doesn’t care about policy. “As you start approaching smaller and smaller technology nodes, things start working very differently,” Patanjali points out.

“The wire that is supposed to be a low resistive element suddenly starts increasing in resistance. And when you pack so many transistors into a very small unit area, the power density starts shooting up. From an archi-